

Supplementary Software Documentation

Companion Code Repository for Technical and Numerical Validation

Manuscript: "Graphene-Assisted Electrostatic Interface Engineering for Radiation-Tolerant MR-DWELL Dosimeters"

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This directory contains the Python computational scripts and analytical models required to reproduce the figures, parametric sweeps, and machine-learning classification workflow presented in the Supplementary Material of the manuscript.

1. Repository Files

File Name	Description & Functional Role
<code>analytic_model.py</code>	Core analytical engine. Computes electrostatic screening factors, quantum capacitance effects, radiation-induced resonance energy shifts, line broadening, and parameter sweeps (barrier thickness, trap density).
<code>plot_fig1.py</code>	Generates Figure S1 : Quantum transmission spectra $T(E)$ across the MR-DWELL heterostructure for various h-BN spacer thicknesses at pre-irradiation ($D = 0$) and post-irradiation ($D = 1$ MGy) states.
<code>plot_fig2.py</code>	Generates Figure S2 : Resonance linewidth evolution $\Gamma(D)$ for the graphene-assisted device in comparison with an unscreened metal-contact benchmark.
<code>classification_model.py</code>	Generates Figure S3 : Synthetic dataset generation and dimensionality reduction workflow (PCA/LDA) for multi-parameter radiation type and dose-level classification.
<code>requirements.txt</code>	Complete list of Python package dependencies with exact version requirements for environmental reproducibility.

2. Installation & Requirements

The code requires Python 3.8 or higher. All necessary library dependencies (including `numpy`, `scipy`, `matplotlib`, and `scikit-learn`) can be installed using `pip`:

```
# Install required Python packages
pip install -r requirements.txt
```

3. Usage & Execution

Each figure script is fully self-contained and can be executed independently from the command line:

```
# Generate Figure S1 (Transmission Spectra)
python plot_fig1.py

# Generate Figure S2 (Linewidth Evolution)
python plot_fig2.py

# Generate Figure S3 (PCA/LDA Classification)
python classification_model.py
```

Upon execution, high-resolution vector and raster graphic files (PNG/PDF) are automatically exported and saved to the current working directory.

4. Important Technical Notes

- **Unified Physics Engine:** `analytic_model.py` acts as the single source of truth for physical constants, geometry, and phenomenological saturation parameters across all plotting routines.
- **Synthetic Classification Workflow:** The classification pipeline in `classification_model.py` generates synthetic feature vectors based on analytical descriptor distributions; it does not require external raw experimental data files.
- **Reproducibility:** All stochastic processes (such as synthetic feature sampling and cross-validation splitting) utilize a fixed random seed (`random_state = 42`) to guarantee exact numerical reproducibility.

Software Versioning & Calibration

All analytical routines correspond to version 2.0 of the calibrated model, incorporating independent phenomenological fitting parameters for low-dose sensitivity (α_{eff}) and asymptotic saturation shift (ΔE_{max}).

5. Citation

If you utilize this code, analytical model, or generated synthetic datasets in your research, please cite the original article:

B. Dosymova and M. Dolgoplov, "Graphene-Assisted Electrostatic Interface Engineering for Radiation-Tolerant MR-DWELL Dosimeters" (2026).